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$z^4 + 4(1-i)z^2 + 3 - 4i = 0$

$w = z^2$

$w^2 + 4(1-i)w + 3 - 4i = 0$

$D = 16(1-i)^2 - 4(3-4i)$

$(1-i)^2 = -2i$

$D = -32i - 12 + 16i = -12 - 16i$

$(a+bi)^2 = -12 - 16i$

$a^2 - b^2 = -12,\ 2ab = -16$

$(a,b) = (2,-4),\ (-2,4)$

$\sqrt{D} = \pm(2-4i)$

$$w = \frac{-4(1-i) \pm (2-4i)}{2}$$

$-4(1-i) = -4 + 4i$

$$w_1 = \frac{(-4+4i) + (2-4i)}{2} = -1$$

$$w_2 = \frac{(-4+4i) - (2-4i)}{2} = -3 + 4i$$

$z^2 = -1 \Rightarrow z = \pm i$

$z^2 = -3 + 4i$

$a^2 - b^2 = -3,\ 2ab = 4$

$(a,b) = (1,2),\ (-1,-2)$

$z = \pm(1+2i)$

$\boxed{z \in \{i, -i, 1+2i, -1-2i\}}$

References